ERICK REINALDO FLORES ZAMBRANO

Economist and AI Engineer | Machine Learning | LLM Systems

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PROFESSIONAL SUMMARY

Economist and AI/ML Engineer with dual academic formation, specialized in end-to-end intelligent systems: predictive models (PyTorch, TensorFlow, scikit-learn), LLM-powered agents, and API-driven automation. Developed FCH-ARX V4, a novel hash algorithm that passed NIST SAC validation, and an autonomous 24/7 job-hunting agent deployed on cloud. Seeking a remote ML/AI engineering role.

KEY PROJECTS AND PORTFOLIO

FCH-ARX V4 - Original Cryptographic Algorithm **[Python, C, NIST FIPS 180-4]**

- Novel hash algorithm from scratch using ARX (Add-Rotate-XOR) with 26 rounds. Passed SAC-NIST: 49.95 pct avalanche (SHA-256 ref: 50.02 pct). Speed: 185 MB/s. Complexity: 2^256.

AI Job Hunter Bot V7 - Autonomous 24/7 Agent **[Python, LinkedIn API, Gmail API, OpenRouter, GitHub Actions]**

- Full automation: scraping 9+ job platforms, LLM cover letters, Gmail with CV, anti-duplicate memory, 24/7 cloud deployment. This CV was sent using this bot.

Algorithmic Trading with Deep Learning **[PyTorch, LSTM, Backtrader, MLflow]**

- LSTM model for SP500: 68 pct directional accuracy. Full pipeline with backtesting and MLflow tracking.

Multi-Agent Financial Analysis System (MCP) **[Python, FastAPI, Docker, LLMs, GARCH]**

- Distributed agents: data scraping (Alpha Vantage), GARCH volatility models, LLM report generation.

RESTful ML Inference API **[FastAPI, Docker, Random Forest, GitHub Actions]**

- Credit risk classification service: Docker, Swagger docs, pytest, CI/CD with GitHub Actions.

TECHNICAL SKILLS

AI / ML: Python (expert), scikit-learn, PyTorch, TensorFlow, LLMs, Prompt Engineering, Hugging Face

Data Engineering: SQL (advanced), PostgreSQL, ETL Pipelines, Pentaho, Airflow (concepts), Spark (concepts)

MLOps / DevOps: FastAPI, Docker, REST APIs, GitHub Actions CI/CD, MLflow, Git

Finance / Econometrics: Time Series (ARIMA, GARCH), Panel Data, Causal Inference, VaR, Backtesting

Business Intelligence: Power BI (DAX, Power Query), Excel VBA, Data Storytelling, SQL Server

Languages: Spanish (Native), English (B2 — technical reading/writing proficient)

PROFESSIONAL EXPERIENCE

Commercial Advisor - Vamoret S.A. (Palmon Group) | **Nov 2022 - Present**

- Lead a team of 3 people in customer service and operational improvement.
- Designed executive Power BI dashboards with real-time KPIs, RFM segmentation, and DAX forecasting.
- Automated monthly inventory reporting with Excel VBA macros - 60 pct time reduction.
- Applied data analysis for pricing strategies and customer retention decisions.

Commercial Advisor - Hularuss S.A. (PepsiCo) | **Oct 2018 - Mar 2021**

- Managed a portfolio of clients using data-driven strategies (Excel + Power BI).
- Trained new advisors in negotiation techniques and NLP (Neuro-Linguistic Programming).
- Implemented inventory rotation and merchandising optimization at points of sale.

EDUCATION

B.Eng. Data Science & Artificial Intelligence — Universidad de Guayaquil | 2023 – Present (4th semester)

B.A. Economics — Universidad Técnica de Manabí (UTM) | 2021 – Present (7th semester, online)

★ Managing two simultaneous university degrees remotely — demonstrates exceptional time management, self-discipline, and commitment to continuous learning.

CERTIFICATIONS

- Python for Finance & Algorithmic Trading — Udemy (40h)
- SQL for Big Data — IBM (40h)
- Deep Learning Specialization — Coursera (in progress)
- LLMs & Prompt Engineering — DeepLearning.AI (in progress)
- Digital Marketing & Web Analytics — Google Activate (50h)
- Power BI for Data Analysis — Universidad de Guayaquil